out) would have gradually gained the upper hand, and *the era of decelerated spatial expansion would have given way to a new era of accelerated expansion.*

In the late 1990s, such reasoning and an in-depth analysis of the data led both the Perlmutter group and the Schmidt group to suggest that Einstein had not been wrong some eight decades earlier when he introduced a cosmological constant into the gravitational equations. The universe, they suggested, does have a cosmological constant.[23] Its magnitude is not what Einstein proposed, since he was chasing a static universe in which gravitational attraction and repulsion matched precisely, and these researchers found that for billions of years repulsion has dominated. But that detail notwithstanding, should the discovery of these groups continue to hold up under the close scrutiny and follow-up studies now under way, Einstein will have once again seen through to a fundamental feature of the universe, one that this time took more than eighty years to be confirmed experimentally.

The recession speed of a supernova depends on the difference between the gravitational pull of ordinary matter and the gravitational push of the "dark energy" supplied by the cosmological constant. Taking the amount of matter, both visible and dark, to be about 30 percent of the critical density, the supernova researchers concluded that the accelerated expansion they had observed required an outward push of a cosmological constant whose dark energy contributes about 70 percent of the critical density.

*This is a remarkable number.* If it's correct, then not only does ordinary matter—protons, neutrons, electrons—constitute a paltry 5 percent of the mass/energy of the universe, and not only does some currently unidentified form of dark matter constitute at least *five times* that amount, but also the *majority* of the mass/energy in the universe is contributed by a totally different and rather mysterious form of dark energy that is spread throughout space. If these ideas are right, they dramatically extend the Copernican revolution: not only are we not the center of the universe, but the stuff of which we're made is like flotsam on the cosmic ocean. If protons, neutrons, and electrons had been left out of the grand design, the total mass/energy of the universe would hardly have been diminished.